Inputs for measuring the complete Mathieu census

Whole documents, programs and data

The target is the complete nine-page original by Leemans and Rowley, *Rank three residually connected geometries for M_{22}, revisited* (2010). The replacement proves the corrected classification of 81 rank-two and 431 rank-three classes and supplies a disposition for every one of the 431 physical source rows. These are classes under the specified group-automorphism and type-reordering equivalence. The unavailable historical `geo[]` numbering is not reconstructed.

Complete written boundary

The detached common paper supplies the full finite-group coset realization, recognition and exhaustive augmentation proofs. The application supplies every Mathieu-specific argument, the full class table and the full literal-source table. The latter two documents are mathematical output, not optional diagnostics. Titles, introductions, repeated table headings and references all remain in the measured text.

Complete document	Pages	Raw NFC nonspace characters
2010 original	9	13,736
Common proof	2	5,516
Mathieu application body	5	13,178
All replacement class tables	10	9,480
All physical-source-row dispositions	9	13,204
All seven mathematical programs	11	31,244

The application's fifth page contains references and is counted in full. The two-page common paper explicitly assumes finite G: equality of cardinalities identifies a completion orbit with the whole finite completion set. This is a recorded scope correction in the modular migration, not a claim that the generic statement was literally unchanged.

The primary written comparison, when separately authorized, will use the whole original against the common paper and all three application documents. A conditional application comparison must first acquire the *whole* common paper, all seven programs, both required data files and the declared external premises. No previous umbrella or other application is an unpaid prerequisite.

Status

This packet supplies measurement inputs and reproducible counts only. It contains no compression factor, admission, historical-priority claim, new result or formal-verification claim. Scientific acceptance and modular migration are bound separately; a distinct numerical input review is still required.

Complete computational boundary

All seven programs are required mathematical computation: exhaustive census, witness validation, geometric coordinates, labelled-code completions, literal configurations, Borel comparisons and complete reconciliation. Their full raw files total 34,695 bytes. The eleven-page appendix prints each file without line pruning. Printed and raw programs are alternative representations, never added together.

Both historical inputs are required in full: the 60,036-byte physical-table contract and the 23,929-byte predecessor-notation JSON. They have no complete printed counterpart here and are additional raw-data terms in either computation-inclusive route. No metadata or unused comparison field is pruned to improve a result.

Three auxiliary scripts total 20,542 bytes and are counted separately. Ten complete generated files total 1,567,120 bytes, including permutation witnesses, labelled-code data, signature tables and reconciliation results. They have separate raw coordinates. Some feed later stages of the same fresh pipeline; they are not charged a second time as independent external inputs. The output inventory remains available for a complete evidence-cost sensitivity. No GAP or mathematical-program replay is performed by this measurement packet.

Extraction and visible symbols

Every physical page is extracted with pypdf 6.13.0 and joined by a newline. The primary coordinate is the untrimmed text in NFC. Sensitivities remove only exact isolated page numbers and the repeated journal footer. All five established units are supplied in NFC and NFKC for raw, furniture-selected and calibrated-selected text: 180 document cells in total. The 22 whole raw files supply a further 220 cells.

Actual font dictionaries, encodings, byte events and embedded outlines are bound. Six observed script/Fraktur classes receive 472 occurrence-specific replacements in the calibrated sensitivity. Raw text stays unchanged. The 27 exceptional and supplementary symbol outlines were visually checked; every used replacement Unicode glyph is bound to its embedded outline. This is not a proof of arbitrary graphical equivalence. The MOG array and table layout remain visible, while their complete graphical complexity is unknown rather than zero.

Retained scope and limits

The 48-page Kılıç–Rowley predecessor is a separate retained notation source. Its targeted defining pages were read; this packet neither counts it as a second recovered original nor silently claims a new full proof of that paper. The complete TomLib catalogue, named group model, exact GAP, SmallGroups, Design and GRAPE algorithms and classical Steiner, Golay, Mathieu and Atlas identifications remain explicit premises. Their full proof and catalogue-construction costs are unknown. Selected software/data hashes are inventory evidence, not a complete source-code cost. There are zero Lean declarations in this census edition.